Project Work: Overall Guideline

Course: Digital System Design
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Semester: Spring 2025-26
Project Duration: 4 Weeks (+2 wks if team shows acceptable progress)
Team Structure: 11 groups × 4 people per group (A, B, C, D)
Deliverable: RTL + Testbench + Verification report for assigned DRS module

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1. Overview

This document provides a **standard work guideline** for each 4-person group working on a digital core module project+. All 11 groups will follow the same workflow, with roles A, B, C, D as defined below.

Each group will implement **one DRS (Design Requirement Specification) module.**

(Note: Some modules assigned to two groups for redundancy.)

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2. Role Definitions (Fixed for 4 Weeks)

Role	Person	Primary Responsibilities
A	Group member 1	RTL Design (core logic), Synthesis, Integration Lead (Week 1)
B	Group member 2	RTL Design (FSM/control), Functional Verification, Post-syn Verification, Integration Lead (Week 2)
C	Group member 3	Testbench Design, Synthesis (+timing), Reporting, Integration Lead (Week 3)
D	Group member 4	Testbench Design, Functional Verification, Post-syn Verification, Reporting, Integration Lead (Week 4)

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3. Task Assignment Matrix

Activity	A	B	C	D
Read DRS specification	✓	✓	✓	✓
RTL Design (core logic)	✓	✓		
Testbench Design			✓	✓
Functional Verification	✓	✓	✓	✓
Synthesis (area + timing)	✓		✓	
Post-synthesis Verification		✓		✓
Reporting			✓	✓
Integration Lead (weekly rotation)	Week 1	Week 2	Week 3	Week 4

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4. Weekly Schedule (4 Weeks)

Week 1: Specification & RTL Design

Activity	Responsible
Read DRS, clarify questions with system team	A, B, C, D
Define internal interfaces, block partitioning	A, B
RTL coding (core logic, FSM, datapath)	A, B
Testbench architecture planning (skeleton)	C, D
RTL first complete, compile clean	A, B
Testbench skeleton ready (interface, clock, reset)	C, D
Integration Lead (Week 1) reports progress	A

Deliverables end of Week 1:

- RTL code (compilation clean)
- Testbench skeleton
- Progress report from Integration Lead A

Week 2: Testbench & Basic Verification

Activity	Responsible
Complete testbench (drivers, monitors)	C, D
Run basic sanity tests (nominal case)	C, D
Functional verification (all test cases from DRS)	A, B, C, D

Activity	Responsible
Debug RTL issues found during verification	A, B
Integration Lead (Week 2) reports progress	B

Deliverables end of Week 2:

- Complete testbench
- Functional verification passed (all test cases)
- Bug report (if any)
- Progress report from Integration Lead B

Week 3: Synthesis & Timing Closure

Activity	Responsible
Run synthesis (area report, timing report)	A, C
Analyze synthesis results, address warnings	A, C
Optimize RTL if timing/area not met	A, B
Re-run synthesis after optimization	A, C
Run post-synthesis gate-level simulation	B, D
Compare pre-syn and post-syn results	B, D
Debug any mismatches (gate sim vs RTL sim)	A, B, C, D
Integration Lead (Week 3) reports progress	C

Deliverables end of Week 3:

- Synthesised netlist
- Gate-level simulation passing
- Progress report from Integration Lead C

Week 4: Integration & Reporting

Day	Activity	Responsible
Day 1–2	Cross-group integration (if module interfaces with others)	A, B, C, D
Day 1–2	Run top-level simulations with neighboring modules	A, B, C, D
Day 3	Prepare final report (see Section 6 for template)	C, D
Day 3	Prepare code package for system integration	A, B
Day 4	Final review, sign-off	All

Day	Activity	Responsible
Day 5	Integration Lead (Week 4) submits final package	D

Deliverables end of Week 4:

- Final RTL code
- Final testbench
- Synthesis reports (area, timing)
- Gate-level simulation logs
- Final report (template below)
- Code package submitted

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5. For Modules with Two Groups (Redundancy)

For modules assigned to **two groups** (Fractional-N Divider, Sigma-Delta Modulator, OTP Controller, Dual-RO Temp Engine), the following **cross-group verification** steps are required:

Week	Activity	Groups Involved
Week 2	Exchange testbenches; run each other's tests	Group X, Group Y
Week 2	Compare simulation results; any mismatch → joint debug	Group X, Group Y
Week 3	Exchange netlists; run cross-synthesis comparison	Group X, Group Y
Week 4	Joint report: document differences and final selection	Group X, Group Y

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6. Report Template (End of Week 4)

Each group must submit a **final report** with the following sections:

```
# Module Report – [Module Name]

**Group ID:** [G1–G11]
**Members:** A: [Name], B: [Name], C: [Name], D: [Name]
**Date:** [Date]

## 1. Executive Summary
- Module implemented: [Name]
- DRS version: [v1.0]
- Status: [Complete / Partial / Issues]

## 2. RTL Implementation
- Total lines of code: [number]
- Finite state machines: [number and description]
- Key design decisions: [brief]
```

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## 3. Testbench and Verification
- Number of test cases: [number]
- Coverage achieved: [% statement / branch / FSM]
- Verification methodology: [directed / constrained random / both]

## 4. Synthesis Results
- Technology node: [65 nm]
- Area: [µm² or gate count]
- Max frequency: [MHz]
- Critical path: [description]

## 5. Gate-Level Simulation
- Pre-syn vs post-syn matching: [Pass / Fail]
- Issues found: [list, if any]

## 6. Integration Notes
- Interfaces to other modules: [list with port names]
- Assumptions made: [list]

## 7. Issues and Resolutions
| Issue | Resolution | Status |
|-------|------------|--------|
| ...   | ...        | Closed |

## 8. Recommendations
- [For system integration]
- [For future improvements]

## 9. Deliverables Checklist
- [ ] RTL code (.v files)
- [ ] Testbench code (.v files)
- [ ] Synthesis script (.tcl)
- [ ] Synthesis report (area, timing)
- [ ] Gate-sim log
- [ ] This report

**Signed:**
A: ________   B: ________
C: ________   D: ________
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7. Communication Guidelines

Topic	Action
Daily standup (within group)	15 min, A leads Week 1, B Week 2, C Week 3, D Week 4
Cross-group coordination	Integration Lead meets with other Integration Leads daily (15 min)
Escalation	Any blocking issue → email to system team within 4 hours
Code repository	Use shared repo; commit at least daily with meaningful messages

8. Success Criteria

Phase	Must Pass
Week 1	RTL compiles without errors
Week 2	All DRS test cases pass in RTL simulation
Week 3	Synthesis meets timing target (TBC by system team)
Week 3	Gate sim matches RTL sim
Week 4	Report delivered, code packaged

9. Appendix: Module to Group Assignment

Module	Group(s)	Redundancy
Fractional-N Divider	G1, G2	2 groups
Sigma-Delta Modulator	G3, G4	2 groups
Frequency Model	G5	1 group
rRO Aging Engine	G6	1 group
OTP Controller	G7, G8	2 groups
Dual-RO Temp Engine	G9, G10	2 groups
Temperature Mapping	G11	1 group